

NALIN JOSHI

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SUMMARY

Analytical Math-Computer Science student focused on stochastic processes and statistical modeling within machine learning. Skilled in applying probabilistic modeling and optimization-driven learning to real-world datasets. Skilled in applying probabilistic modeling and optimization-driven learning to real-world datasets. Experienced in developing interpretable models and supporting machine learning pipelines, experimentation, and deployment-ready systems.

EDUCATION

University of California, San Diego

Bachelor of Science in Mathematics-Computer Science

Expected in 06/2026

- **Relevant Coursework:** Stochastic Processes, Regression Analysis, Linear Algebra, Multivariate Calculus, Discrete Mathematics, Data Structures and Algorithms

TECHNICAL SKILLS

Quantitative & Mathematical Foundations: Bayesian Inference, Probability Theory, Continuous/Discrete time Markov chains, Brownian Motion, Group/Ring theory, Graph Theory

Machine Learning & Optimization: Probabilistic modeling, Bayesian estimation (MLE/MAP), Likelihood maximization, Regularization, Preference and utility modeling; Uncertainty quantification, Entropy

Programming (Proficient): Python, Java, C/C++, Django/REST APIs, Git and version control, PyTorch, TensorBoard, Data visualization, Backend data pipelines, Experiment logging and monitoring

RELEVANT EXPERIENCE

Value Alignment for Customer Behavior Analysis

October 2025 - December 2025

- Built a probabilistic framework to infer latent consumer preferences from behavioral data using random utility modeling.
- Modeled decision-making with multinomial logit and Bayesian hierarchical models to capture preference heterogeneity and uncertainty.
- Estimated rationality parameters (λ) and defined alignment metrics using regret-based and divergence-based measures to quantify decision consistency.
- Implemented Bayesian inference in PyMC using HMC/NUTS and evaluated robustness under contextual shifts (time, traffic, seasonality).

Instruction-Tuned LLMs for Academic Summarization | RapidFire AI

December 2025 - January 2026

- Fine-tuned sequence-to-sequence transformer models for academic paper summarization using parameter-efficient methods (LoRA).
- Compared base models, prompt-engineered approaches, and LoRA-fine-tuned models through controlled evaluations to assess performance tradeoffs.
- Designed evaluation pipelines using ROUGE-style metrics, qualitative error analysis, and loss-based training diagnostics.
- Implemented training workflows with Hugging Face datasets, model checkpoints, and experiment tracking to support iterative model development.

UCSD Machine Learning Certification

August 2024 - February 2025

- Designed and implemented end-to-end data analysis workflows to clean, transform, and visualize large datasets using Python.
- Engineered features and prepared modeling datasets through encoding, binning, normalization, and dimensionality reduction.
- Evaluated supervised learning models for mini projects including linear and logistic regression, decision trees, random forests, and gradient boosting.
- Applied unsupervised learning and statistical inference methods, including K-means, PCA, chi-squared tests, ANOVA, and t-tests, to support data-driven analysis and validation.

Backend Software Development Intern | GoLEAD

January 2024 - April 2024

- Developed RESTful APIs using Django REST Framework to support backend functionality, including a student engagement ranking system.
- Integrated OpenAI APIs into an LMS platform to enable AI-driven features, including a PDF-based chatbot for course materials.
- Built and deployed an interactive Streamlit application to demonstrate LLM and API integrations, delivering an MVP to stakeholders.
- Improved demo usability by integrating Streamlit into an existing codebase to enhance UI/UX and interaction flow.
- Designed and executed API test suites and incorporated testing into development workflows to support reliability.
- Containerized backend services using Docker and authored Dockerfiles to manage dependencies and deployment.
- Used Git and GitHub for version control, including branching, feature isolation, and merge conflict resolution.

CERTIFICATIONS

- UCSD Extended Studies Machine Learning Certification